

Preserving categoricity and complexity of relations

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Abstract

Work of Goncharov [6], by himself, and with other collaborators [10], [4], shows that for each computable ordinal α , there is a structure that is Δ_α^0 categorical but not relatively Δ_α^0 categorical. The original examples were not familiar algebraic kinds of structures. Hirschfeldt, Khoushainov, Shore, and Slinko [12], showed that for $\alpha = 1$, there are further examples in several familiar classes of structures, including rings and 2-step nilpotent groups. Tussupov [20], [21] showed the same for all computable successor ordinals. Here we extend to computable limit ordinals. Hirschfeldt, Kramer, R. Miller, and Shlapentokh [13] produced an algebraic field that is computably categorical but not relatively computably categorical. Here we show that for each computable limit ordinal $\alpha > \omega$, there is a field which is Δ_α^0 categorical but not relatively Δ_α^0 categorical. The earlier papers give examples on dimension and complexity of relations, and we give results of these kinds as well.

1 Introduction

Here are the definitions of Δ_α^0 categoricity and relative Δ_α^0 categoricity, from [1].

Definition 1. *Let $\mathcal{A}$ be a computable structure.*

1. $\mathcal{A}$ is Δ_α^0 categorical if for all computable copies $\mathcal{B}$, there is a Δ_α^0 isomorphism from $\mathcal{A}$ onto $\mathcal{B}$.
2. $\mathcal{A}$ is relatively Δ_α^0 categorical if for all copies $\mathcal{B}$, there is an isomorphism from $\mathcal{A}$ onto $\mathcal{B}$ that is $\Delta_\alpha^0(\mathcal{B})$.

Relative Δ_α^0 categoricity is nicer than Δ_α^0 categoricity. It is truly intrinsic; it results from something definable in the structure see [1], [3].

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Theorem 1.1 (Ash-K-Manasse-Slaman, Chisholm). *For a computable structure $\mathcal{A}$, the following are equivalent:*

1. $\mathcal{A}$ is relatively Δ_α^0 categorical,
2. there is a “formally Σ_α^0 Scott family” for $\mathcal{A}$, where this is a c.e. set Ψ of computable Σ_α formulas, with fixed parameters, such that
 - each tuple $\bar{a}$ satisfies some $\psi \in \Psi$,
 - if tuples $\bar{a}, \bar{a}'$ satisfy the same formula $\psi \in \Psi$, there is an automorphism of $\mathcal{A}$ taking $\bar{a}$ to $\bar{a}'$.

In computable structure theory, it is natural to focus on computable structures. Thus, it would be pleasing if Δ_α^0 categoricity and relative Δ_α^0 categoricity were the same. Sadly, this is not the case. The first example is due to Goncharov [6].

Theorem 1.2 (Goncharov). *There is a structure $\mathcal{A}$ (a graph) that is computably categorical but not relatively computably categorical.*

Idea of Proof. The example is based on a result of Badaev and Selivanov [2], [17] in enumeration theory. The enumeration result concerns “Friedberg” enumerations of a family of sets; these are enumerations in which each set has just one index. The resulting structure is rigid.

□

Goncharov’s basic example lifts to successor ordinals [10].

Theorem 1.3 (Goncharov-Harizanov-K-McCoy-R. Miller-Solomon). *For each computable successor ordinal α , there is a structure $\mathcal{A}$ that is Δ_α^0 categorical but not relatively Δ_α^0 categorical.*

Idea of Proof. The structure is obtained by relativizing the results of Badaev and Selivanov and coding the resulting Δ_α^0 structure in a computable one. Again the resulting structure is rigid.

□

For limit ordinals, there are examples in [4].

Theorem 1.4 (Chisholm-Fokina-Goncharov-Harizanov-K-Quinn). *For each computable limit ordinal α , there is a structure $\mathcal{A}$ that is Δ_α^0 categorical but not relatively Δ_α^0 categorical.*

Idea of proof. The proof involves first giving a new result in enumeration theory. What is enumerated is a family of functions, where a prescribed oracle $\Delta_{\alpha_n}^0$ gives the value of the functions at n . The enumerations are “anti-Friedberg”; i.e., each function has infinitely many indices. As a consequence, the structures we obtain are not rigid.

□

Goncharov's original example, and the other examples mentioned so far, are not algebraic. We might hope that for the classes of algebraic structures that we care most about, computable categoricity and relative computable categoricity coincide. Goncharov proved results saying exactly this for several classes [7], [8],[9].

Theorem 1.5 (Goncharov). *For $\mathcal{A}$ in any of the following classes, if $\mathcal{A}$ is computably categorical, then it is relatively computably categorical.*

1. *linear orderings*
2. *Boolean algebras*
3. *Abelian groups*

These results of Goncharov are very satisfying. For each class, he gave a structural characterization of the computably categorical members. Among linear orderings, they are the ones with just finitely many successor pairs. Among Boolean algebras, they are the ones with just finitely many atoms.

Hirschfeldt, Khoushainov, Shore, and Slinko [12] showed that there are algebraic structures which are computably categorical but not relatively computably categorical.

Theorem 1.6 (Hirschfeldt-Khoushainov-Shore-Slinko). *In each of the following classes, there is a structure that is computably categorical and not relatively computably categorical:*

1. *undirected graphs*
2. *partial orderings*
3. *lattices*
4. *rings*
5. *2-step nilpotent groups*

To prove this, and several other results, Hirschfeldt, Khoushainov, Shore, and Slinko took an abstract approach. They gave conditions sufficient to transfer various properties from a directed graph $\mathcal{G}^*$ to a structure $\mathcal{A}^*$ —computable categoricity, computable dimension, etc. Tussupov [20] produced, for each computable successor ordinal α , in each of the classes above, a structure that is Δ_α^0 categorical but not relatively Δ_α^0 categorical. The conditions guaranteed that if the input graph was rigid, the output structure is also rigid.

In the present paper, we extend the result to limit ordinals. We use an abstract approach. Our conditions are weaker than those of [12]. We give separate results for transferring Δ_α^0 categoricity from an input structure to an output structure and for transferring relative Δ_α^0 categoricity from the output

structure to the input structure. This is useful for our result on fields. In addition, our output structures need not be rigid even when the input structure is rigid. The primary focus of [12] was computable dimension, with computable categoricity as a special case. There were also results on complexity of relations. Our primary focus is Δ_α^0 categoricity, but we also obtain results on dimension and on complexity of relations.

In Section 2, we first give the general conditions of [12] for the case where a copy of the input graph is definable in the output structure. We then give our conditions for the same case, with applications. In Section 3, we give the general conditions of [12] for the case where a copy of the input graph is obtained as a definable quotient in the output structure, and we give our conditions for this case, with further applications. In Sections 2 and 3, we include results on dimension. In Section 4, we give our result on fields. In Section 5, we give conditions for transferring intrinsically Σ_α^0 relations forward and relatively intrinsically Σ_α^0 relations back.

2 Transfer of categoricity—first case

We first give the conditions of [12] for the case where the input structure is definable in the output structure. After that, we give our results for this same case, with weaker conditions. We can apply the results to obtain, for each computable ordinal $\alpha \geq 1$, a directed graph, a partial ordering, and a lattice which are Δ_α^0 categorical but not relatively Δ_α^0 categorical.

2.1 Conditions of Hirschfeldt et al—first case

Here are the conditions for the first general result in [12], guaranteeing transfer of computable dimension and other properties. Let $\mathcal{G}^*$ be a directed graph, and let $\mathcal{B}^*$ be a structure for a finite relational language satisfying the following conditions:

1. there is some $\mathcal{G}_{\mathcal{B}^*} \cong \mathcal{G}^*$ defined in $\mathcal{B}^*$ by computable Σ_1 and computable Π_1 formulas—we allow finitely many parameters, but we require that any automorphism of $\mathcal{B}^*$ leaves invariant the universe and the relations of $\mathcal{G}_{\mathcal{B}^*}$,
2. there is a computable operator Φ , such that for $\mathcal{G} \cong \mathcal{G}^*$, $\Phi(\mathcal{G}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$, and if $\mathcal{G}_{\mathcal{B}}$ is defined in $\mathcal{B}$ by the formulas that defined $\mathcal{G}_{\mathcal{B}^*}$, then $\mathcal{G} \cong_f \mathcal{G}_{\mathcal{B}}$,
3. every automorphism of $\mathcal{G}_{\mathcal{B}}$ extends to an automorphism of $\mathcal{B}$,
4. there is a “formally Σ_1 defining family for $\mathcal{B}^*$ over $\mathcal{G}_{\mathcal{B}^*}$ ”, where this is a c.e. set Ψ of computable Σ_1 formulas, such that for each $a \in \mathcal{B}^*$ (in [12], this does not involve parameters),

- b satisfies $\psi(\bar{g}, x)$ for some $\psi \in \Psi$ and $\bar{g} \in \mathcal{G}_{\mathcal{B}^*}$,

- if b, b' both satisfy $\varphi(\bar{g}, x)$, for $\psi \in \Psi$ and $\bar{g} \in G_{\mathcal{B}^*}$, then $b = b'$.

In [12], Hirschfeldt et al showed that under the conditions above, many properties transfer from $\mathcal{G}^*$ to $\mathcal{B}^*$. In particular, $\mathcal{G}^*$ is computably categorical, or relatively computably categorical if and only if $\mathcal{B}^*$ has this property; $\mathcal{G}^*$ has computable dimension n if and only if $\mathcal{B}^*$ does, etc.

2.2 Transferring Δ_α^0 categoricity—first case

We imitate the abstract approach from [12]. We give, in separate results, conditions for transferring Δ_α^0 categoricity from an input structure to an output structure, and for transferring relative Δ_α^0 categoricity from the output structure to the input structure.

Proposition 2.1. *Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there exists $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, defined in $\mathcal{B}^*$ by formulas $D(x)$, $R_i(\bar{x})$. We allow finitely many parameters in the formulas, but we require that the interpretations are invariant under automorphisms of $\mathcal{B}^*$ (so the parameters are inessential). Suppose there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f is an isomorphism from $\mathcal{A}$ onto the copy $\mathcal{A}_{\mathcal{B}}$ defined in $\mathcal{B}$. If $\mathcal{B}^*$ is relatively Δ_α^0 categorical, then so is $\mathcal{A}^*$.*

Proof. We suppose that $\Phi(\mathcal{A}^*) = (\mathcal{B}^*, f^*)$. Let $\mathcal{A} \cong \mathcal{A}^*$. Let $\Phi(\mathcal{A}) = (\mathcal{B}, f)$. Then $\mathcal{B}^* \cong \mathcal{B}$. There is an isomorphism h from $\mathcal{B}^*$ onto $\mathcal{B}$ such that h is Δ_α^0 relative to $\mathcal{B}$, so it is Δ_α^0 relative to $\mathcal{A}$. The isomorphism h maps $\mathcal{A}_{\mathcal{B}^*}$ isomorphically onto $\mathcal{A}_{\mathcal{B}}$. We obtain an isomorphism k from $\mathcal{A}^*$ onto $\mathcal{A}$, where $k(x) = y$ if $h(f^*(x)) = f(y)$. This k is Δ_α^0 relative to $\mathcal{A}$. Therefore, $\mathcal{A}$ is relatively Δ_α^0 -categorical. □

Remark 1. *The proof above shows more.*

1. *If $\mathcal{B}$ is Δ_α^0 categorical, then $\mathcal{A}$ is also.*
2. *If the Δ_α^0 dimension of $\mathcal{B}$ is at most n , then the same is true of $\mathcal{A}$.*

Proposition 2.2. *Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there is some $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, defined in $\mathcal{B}^*$ by computable Σ_1 and computable Π_1 formulas. We allow finitely many parameters, but we require that the interpretations are invariant under automorphisms of $\mathcal{B}^*$. Suppose Φ is a Turing operator, defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f is an isomorphism from $\mathcal{A}$ onto the copy $\mathcal{A}_{\mathcal{B}}$, such that*

- *every automorphism of $\mathcal{A}_{\mathcal{B}^*}$ extends to an automorphism of $\mathcal{B}^*$, and*
- *there is a “formally Σ_α^0 Scott family for $\mathcal{B}^*$ over $\mathcal{A}_{\mathcal{B}^*}$ ”, where this is a c.e. set Ψ of computable Σ_α formulas such that*

- (a) for each tuple $\bar{b}$ in $\mathcal{B}^*$, there exist a formula $\psi \in \Psi$ and a tuple $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$ such that $\mathcal{B}^* \models \psi(\bar{d}, \bar{b})$,
- (b) if $\bar{b}, \bar{b}'$ both satisfy $\psi(\bar{d}, \bar{x})$, where $\psi \in \Psi$ and $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$, there is an automorphism of $\mathcal{B}^*$ taking $\bar{b}$ to $\bar{b}'$ and fixing $\mathcal{A}_{\mathcal{B}^*}$.

If $\mathcal{A}^*$ is Δ_α^0 -categorical, then so is $\mathcal{B}^*$.

Proof. We may suppose that $\Phi(\mathcal{A}^*) = (\mathcal{B}^*, f^*)$. Let $\mathcal{B}$ be a computable copy of $\mathcal{B}^*$. Let $\mathcal{A}_{\mathcal{B}}$ be the definable copy of $\mathcal{A}^*$ in $\mathcal{B}$. Then $\mathcal{A}_{\mathcal{B}}$ is computable. There is a Δ_α^0 isomorphism g from $\mathcal{A}_{\mathcal{B}^*}$ onto $\mathcal{A}_{\mathcal{B}}$. The fact that every automorphism of $\mathcal{A}_{\mathcal{B}^*}$ extends to an automorphism of $\mathcal{B}^*$ implies that every isomorphism from $\mathcal{A}_{\mathcal{B}^*}$ onto $\mathcal{A}_{\mathcal{B}}$ extends to an isomorphism from $\mathcal{B}^*$ onto $\mathcal{B}$. We want a Δ_α^0 isomorphism $h \supseteq g$. We use Ψ . Let $\mathcal{F}$ be the family of finite partial functions taking a tuple $\bar{d}, \bar{b}$ in $\mathcal{B}^*$ to a tuple $\bar{d}', \bar{b}'$ in $\mathcal{B}$ such that for some formula $\psi(\bar{u}, \bar{x}) \in \Psi$, $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$, $g(\bar{d}) = \bar{d}'$, $\mathcal{B}^* \models \psi(\bar{d}, \bar{b})$ and $\mathcal{B} \models \psi(\bar{d}', \bar{b}')$.

Claim: $\mathcal{F}$ has the back-and-forth property.

Proof of Claim. Consider a function p in $\mathcal{F}$ taking $\bar{d}, \bar{b}$ to $\bar{d}', \bar{b}'$, where $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$. We have $\bar{d}' = g(\bar{d})$, and for some $\psi \in \Psi$, $\mathcal{B} \models \psi(\bar{d}', \bar{b}')$ and $\mathcal{B}^* \models \psi(\bar{d}, \bar{b})$. Take c^* a further element of $\mathcal{B}^*$. We want an extension of p with c^* in the domain. We have $\psi' \in \Psi$ and $\bar{d}_1 \in \mathcal{A}_{\mathcal{B}^*}$ such that $\mathcal{B}^* \models \psi'(\bar{d}, \bar{d}_1, \bar{b}, c^*)$. Let $\bar{d}'_1 = g(\bar{d}_1)$. The partial isomorphism g from $\mathcal{A}_{\mathcal{B}^*}$ to $\mathcal{A}_{\mathcal{B}}$ extends to an isomorphism from $\mathcal{B}^*$ to $\mathcal{B}$, so there exist $\bar{b}'', c$ such that $\mathcal{B} \models \psi'(\bar{d}', \bar{d}'_1, \bar{b}'', c)$ & $\psi(\bar{d}', \bar{b}'')$. Since $\bar{b}'$ and $\bar{b}''$ both satisfy $\psi(\bar{d}', \bar{x})$, there is an automorphism of $\mathcal{B}$ fixing all elements of $\mathcal{A}_{\mathcal{B}}$ and taking $\bar{b}''$ to $\bar{b}'$, and taking c to some c' . Then $\mathcal{B} \models \psi'(\bar{d}', \bar{d}'_1, \bar{b}', c')$, so we have the required extension of p .

Take c' a further element of $\mathcal{B}$. We want an extension of p with c' in the range. We have $\psi' \in \Psi$, $\bar{d}'_1$ in $\mathcal{A}_{\mathcal{B}}$ such that $\mathcal{B} \models \psi'(\bar{d}', \bar{d}'_1, \bar{b}', c')$. We get $\bar{b}^*, c^*$ in $\mathcal{B}^*$ such that $\mathcal{B}^* \models \psi'(\bar{d}, \bar{d}_1, \bar{b}^*, c^*)$ & $\psi(\bar{d}, \bar{b}^*)$. Then there is an automorphism of $\mathcal{B}^*$ fixing the elements of $\mathcal{A}_{\mathcal{B}^*}$, and taking $\bar{b}^*$ to $\bar{b}'$, and c^* to some c . Again we have the required extension of p .

□

□

Remark 2. The proof above shows more.

1. If $\mathcal{A}$ is relatively Δ_α^0 categorical, then so is $\mathcal{B}$.
2. If the Δ_α^0 dimension of $\mathcal{A}$ is at most n , then the same is true of $\mathcal{B}$.

Combining the conditions of our two propositions, we have the following.

Combined conditions. Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there is some $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, where $\mathcal{A}_{\mathcal{B}^*}$ is defined in $\mathcal{B}^*$

by computable Σ_1 and computable Π_1 formulas. There may be a finite tuple of parameters, but the interpretations of the formulas are invariant under automorphisms. Suppose there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f is an isomorphism from $\mathcal{A}$ onto the copy $\mathcal{A}_{\mathcal{B}}$ satisfying the following conditions:

1. every automorphism of $\mathcal{A}_{\mathcal{B}^*}$ extends to an automorphism of $\mathcal{B}^*$,
2. there is a “formally Σ_α^0 Scott family for $\mathcal{B}^*$ over $\mathcal{A}_{\mathcal{B}^*}$ ”, where this is a c.e. set Ψ of computable Σ_α formulas such that
 - (a) for each tuple $\bar{b}$ in $\mathcal{B}^*$, there exist $\psi \in \Psi$ and $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$ such that $\mathcal{B}^* \models \psi(\bar{d}, \bar{b})$,
 - (b) if $\bar{b}, \bar{b}'$ both satisfy $\psi(\bar{d}, \bar{x})$, where $\psi \in \Psi$ and $\bar{d} \in \mathcal{A}_{\mathcal{B}^*}$, there is an automorphism of $\mathcal{B}^*$ taking $\bar{b}$ to $\bar{b}'$ and fixing $\mathcal{A}_{\mathcal{B}^*}$.

Remark 3. *It is easy to see that the conditions of Hirschfeldt et al imply our conditions. Even when combined, our conditions are weaker.*

2.3 Applications of Propositions 2.1 and 2.2

There are familiar schemes for coding a structure for a finite relational language in a graph. We may use either a directed graph or an undirected graph. One coding scheme is described in Marker’s text on model theory [14]. This is what we use in the following result. A similar scheme is in a paper by Nies [16].

Proposition 2.3. *For each structure $\mathcal{A}^*$ for a finite relational language, with universe a subset of ω , there is a directed graph $\mathcal{B}^*$, computable relative to $\mathcal{A}^*$, such that $\mathcal{A}^*$ and $\mathcal{B}^*$ satisfy the conditions of Hirschfeldt et al stated above.*

Sketch of proof. For each $n \in \mathcal{A}^*$, we put into the directed graph $\mathcal{B}^*$ an element b_n . We connect b_n to a triangle (with ordered vertices). For a binary relation R in $\mathcal{A}^*$, we represent each pair $(k, j) \in \mathcal{A}^*$, by a point $c_{(k,j)} \in \mathcal{B}^*$. We connect b_k to $c_{(k,j)}$ directly, and we connect b_j to $c_{(k,j)}$ with one stop. If $R(k, j)$ holds in $\mathcal{A}^*$, we connect $c_{(k,j)}$ to a square, and otherwise, we connect it to a pentagon. In general, for each relation R , we choose a new pair of n -gons (replacing the square and the pentagon). If R has r places, then for each r -tuple $(k_1, \dots, k_r) \in \mathcal{A}^*$, and we have a special point $p_{(k_1, \dots, k_r)}^R$ connected to $k_1, \dots, k_r$ with the appropriate number of stops, and connected to the appropriate n -gon to indicate whether $R(k_1, \dots, k_r)$ holds. □

Corollary 2.4 (Directed graphs). *For each computable ordinal $\alpha \geq 1$ there is a directed graph $\mathcal{G}^*$ that is Δ_α^0 categorical but not relatively Δ_α^0 categorical.*

Proof. For each α , we start with a computable structure $\mathcal{A}^*$ from [6], [10], or [4] such that $\mathcal{A}^*$ is Δ_α^0 categorical but not relatively Δ_α^0 categorical. This $\mathcal{A}^*$ is a structure for a finite relational language. Applying Proposition 2.3, we get a

directed graph $\mathcal{G}^*$ such that $\mathcal{A}^*$ and $\mathcal{G}^*$ satisfy the conditions of Hirschfeldt et al, so they also satisfy the conditions of our Propositions 2.1 and 2.2. Therefore, $\mathcal{G}^*$ is Δ_α^0 categorical and not relatively Δ_α^0 categorical. $\square$

Proposition 2.5 (Partial orderings and lattices). *For each computable ordinal $\alpha \geq 1$, there is a structure in each of the following classes that is Δ_α^0 categorical but not relatively Δ_α^0 categorical:*

1. *partial orderings*
2. *lattices*

Proof. We start with a directed graph $\mathcal{G}^*$ that is Δ_α^0 categorical but not relatively Δ_α^0 categorical. In [12], it is shown that for any directed graph $\mathcal{G}^*$, there is a structure $\mathcal{B}^*$, in each of the classes above, satisfying the conditions of their first general result. Then the input graph $\mathcal{G}^*$ and the output partial ordering, or lattice, $\mathcal{B}^*$ also satisfy the conditions of our two propositions. Therefore, $\mathcal{B}^*$ is Δ_α^0 categorical but not relatively Δ_α^0 categorical. $\square$

3 Transfer of categoricity—second case

We have considered the case where a copy of the input structure is definable in the output structure. Now, we turn to the case where a copy of the input structure is obtained as a definable quotient in the output structure. We first give the conditions of [12] for this case. Next, we give our results for the same case, with weaker conditions. Finally, we apply the results to show that there are rings, integral domains, commutative semigroups, and 2-step nilpotent groups which are Δ_α^0 categorical but not relatively Δ_α^0 categorical.

3.1 Conditions of Hirschfeldt et al—second case

Here are the conditions for the second general result in [12], guaranteeing the transfer of computable dimension and other properties. Let $\mathcal{G}^*$ be a directed graph, and let $\mathcal{B}^*$ be a structure in a finite relational language. Suppose the following conditions hold:

1. there exists a quotient structure $\mathcal{G}_{\mathcal{B}^*} \cong \mathcal{G}^*$ defined in $\mathcal{B}^*$ by computable Σ_1 and computable Π_1 formulas—we allow finitely many parameters in the defining formulas $D(x)$, $\sim$, and $R(x, y)$, but the interpretations are invariant under automorphisms,
2. there is a computable operator Φ , such that for $\mathcal{G} \cong \mathcal{G}^*$, $\Phi(\mathcal{G}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$, $\text{ran}(f)$ is a choice set for $\sim$, with exactly one element from each $\sim$ -class, and f induces an isomorphism from $\mathcal{G}$ onto $\mathcal{G}_{\mathcal{B}}$,

3. for any function g mapping one choice set for $\sim$ to another, if g induces an automorphism $\mathcal{G}_{\mathcal{B}^*}$, then g extends to an automorphism of $\mathcal{B}^*$,
4. there is a formally Σ_1 defining family for $\mathcal{B}^*$ over $\text{ran}(f)$, where f is the second component of $\Phi(\mathcal{G}^*)$.

Hirschfeldt et al showed that under the conditions above, many properties transfer from $\mathcal{G}^*$ to $\mathcal{B}^*$. In particular, $\mathcal{G}^*$ is computably categorical, or relatively computably categorical, or has computable dimension n , if and only if the same is true of $\mathcal{B}^*$.

3.2 Transferring Δ_α^0 categoricity—second case

We give, in separate results, conditions for transferring Δ_α^0 categoricity from an input structure to an output structure, and for transferring relative Δ_α^0 categoricity from the output structure to the input structure.

Proposition 3.1. *Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there is a quotient structure $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, definable in $\mathcal{B}^*$. The defining formulas $D(x)$, $\sim$, and $R_i(\bar{x})$ may have finitely many parameters, but the interpretations of the relations must be invariant under automorphisms. Suppose that the formula $\sim$ is computable Σ_α . Finally, suppose that there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f induces an isomorphism from $\mathcal{A}$ onto the quotient structure $\mathcal{A}_{\mathcal{B}}$. If $\mathcal{B}^*$ is relatively Δ_α^0 categorical, then so is $\mathcal{A}^*$.*

Proof. We may suppose $\Phi(\mathcal{A}^*) = (\mathcal{B}^*, f^*)$. Let $\mathcal{A} \cong \mathcal{A}^*$, and let $\Phi(\mathcal{A}) = (\mathcal{B}, f)$. There is an isomorphism h from $\mathcal{B}^*$ onto $\mathcal{B}$ that is Δ_α^0 relative to $\mathcal{B}$, so it is Δ_α^0 relative to $\mathcal{A}$. Let $\bar{b}$ be the tuple of parameters used in the formulas $D(x)$, $\sim$, and $R_i(\bar{x})$. With the parameters $h(\bar{b})$, these formulas define the quotient structure $\mathcal{A}_{\mathcal{B}}$, and f induces an isomorphism from $\mathcal{A}$ onto $\mathcal{A}_{\mathcal{B}}$. Let g map $\mathcal{A}^*$ to $\mathcal{A}$, where $g(a) = a'$ if $(\exists b, b') (h \circ f^*(a) = b \ \& \ f(a') = b' \ \& \ b \sim b')$. Then g is an isomorphism that is Δ_α^0 relative to $\mathcal{A}$. □

Proposition 3.2. *Let $\mathcal{A}^*, \mathcal{B}^*$ be structures for finite relational languages. Suppose there are computable Σ_1 and computable Π_1 formulas defining a quotient $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$. The formulas may have parameters, but the interpretations are invariant under automorphisms. Suppose there is a Turing operator Φ such that for all $\mathcal{A} \cong \mathcal{A}^*$, $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f induces an isomorphism from $\mathcal{A}$ onto $\mathcal{A}_{\mathcal{B}}$. Suppose*

- for any g mapping one choice set to another, if g induces an automorphism of $\mathcal{A}_{\mathcal{B}^*}$, then g extends to an automorphism of $\mathcal{B}^*$,
- there is a “formally Σ_α^0 Scott family” for $\mathcal{B}^*$ over $\text{ran}(f)$, where this is a c.e. set Ψ of computable Σ_α formulas such that

(a) for each $\bar{b} \in \mathcal{B}^*$, there is some $\bar{d} \in \text{ran}(f)$ such that $\mathcal{B} \models \psi(\bar{d}, \bar{b})$,

- (b) if $\bar{b}, \bar{b}'$ satisfy the same $\psi(\bar{d}, \bar{x})$, then there is an automorphism of $\mathcal{B}^*$ taking $\bar{b}'$ to $\bar{b}$ and fixing $\text{ran}(f)$.

If $\mathcal{A}^*$ is Δ_α^0 categorical, so is $\mathcal{B}^*$.

Proof. We suppose that $\Phi(\mathcal{A}^*) = (\mathcal{B}^*, f^*)$. Let $\mathcal{B}$ be a computable copy of $\mathcal{B}^*$. We must show that there is a Δ_α^0 isomorphism from $\mathcal{B}^*$ onto $\mathcal{B}$. In $\mathcal{B}$, the interpretations of the formulas $D(x), \sim, R_i$ that give the definable quotient $\mathcal{A}_\mathcal{B}$ are all computable relative to $\mathcal{B}$. We obtain $\mathcal{A}' \cong \mathcal{A}^*$, where the universe of $\mathcal{A}'$ consists of the elements that satisfy $D(x)$ and are first in their $\sim$ class, where by first we mean under the order inherited by ω , and the relations are the restrictions of the $R_i(\bar{x})$. Then $\mathcal{A}'$ is computable in $\mathcal{B}$. We have an isomorphism g from $\mathcal{A}^*$ onto $\mathcal{A}'$, where g is Δ_α^0 relative to $\mathcal{B}$. Let h be the partial function taking $\text{ran}(f^*)$ onto $\mathcal{A}'$, where $h(x) = g(f^{-1}(x))$. To extend h to the rest of $\mathcal{B}^*$, we use Ψ . Let $\mathcal{F}$ be the set of finite functions taking a tuple $\bar{b} \in \mathcal{B}^*$ to a tuple $\bar{b}' \in \mathcal{B}$ such that for some $\psi(\bar{u}, \bar{x}) \in \Psi$ and some $\bar{d} \in \text{ran}(f)$ and the corresponding $\bar{d}' = h(\bar{d})$, we have $\mathcal{B} \models \psi(\bar{d}, \bar{b})$ and $\mathcal{B}' \models \psi(\bar{d}', \bar{b}')$.

Claim: $\mathcal{F}$ has the back-and-forth property.

Proof of Claim. Suppose $p \in \mathcal{F}$ takes $\bar{b}^*$ to $\bar{b}$, where for some $\psi \in \Psi$ and $\bar{d} \in \text{ran}(p)$, with $\bar{d}' = h(\bar{d})$, we have $\mathcal{B}^* \models \psi(\bar{d}, \bar{b}^*)$ and $\mathcal{B} \models \psi(\bar{d}', \bar{b})$. Let c^* be a further element of $\mathcal{B}^*$. There exist $\psi' \in \Psi$ and $\bar{d}_1 \in \text{ran}(p)$ such that $\mathcal{B}^* \models \psi'(\bar{d}_1, \bar{b}^*, c^*)$. Let $\bar{d}'_1 = h(\bar{d}_1)$. The function h , defined on $\text{ran}(p)$, extends to an isomorphism k from $\mathcal{B}^*$ onto $\mathcal{B}$. Say k takes $\bar{b}^*$ to $\bar{b}'$ and c^* to c . Then $\mathcal{B} \models \psi(\bar{d}', \bar{b}')$ & $\psi'(\bar{d}'_1, \bar{b}', c)$. There is an automorphism of $\mathcal{B}$ that fixes $\text{ran}(h)$, takes $\bar{b}'$ to $\bar{b}$, and takes c to some c' . Then $\mathcal{B} \models \psi'(\bar{d}'_1, \bar{b}, c')$, so we have our extension of p with c^* in the domain. Similarly, if c is a further element of $\mathcal{B}$, we can find an extension of p with c in the range. □

□

We combine the conditions of our Propositions 3.1 and 3.2.

Combined conditions. Let $\mathcal{A}^*, \mathcal{B}^*$ be structures for finite relational languages. Suppose there is a structure $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$ such that $\mathcal{A}_{\mathcal{B}^*}$ is a definable quotient in $\mathcal{B}^*$, defining formulas $D(x), \sim(x, y)$, and $R_i(\bar{x})$ that are both computable Σ_1 and computable Π_1 . The defining formulas may have finitely many parameters, but the interpretations must be invariant under automorphisms of $\mathcal{B}^*$. Suppose there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f induces an isomorphism from $\mathcal{A}$ onto $\mathcal{A}_\mathcal{B}$. Suppose

1. for any g mapping one choice set for $\sim$ to another, if g induces an automorphism of $\mathcal{A}_{\mathcal{B}^*}$, then g extends to an automorphism of $\mathcal{B}^*$,

2. there is a “formally Σ_α^0 Scott family for $\mathcal{B}^*$ over $\text{ran}(f)$ ”, where this is a c.e. set Ψ of computable Σ_α formulas such that
 - (a) for each $\bar{b} \in \mathcal{B}^*$, there is some $\bar{d} \in \text{ran}(f)$ such that $\mathcal{B} \models \psi(\bar{d}, \bar{b})$,
 - (a) if $\bar{b}, \bar{b}'$ satisfy the same $\psi(\bar{d}, \bar{x})$, then there is an automorphism of $\mathcal{B}^*$ taking $\bar{b}'$ to $\bar{b}$ and fixing $\text{ran}(f)$.

Remark 4. *It is easy to see that the conditions of Hirschfeldt et al, for this second case, imply our conditions.*

3.3 Applications of Propositions 3.1 and 3.2

We can apply the results of Section 3 to obtain algebraic examples.

Proposition 3.3 (Rings, integral domains, commutative semigroups, 2-step nilpotent groups). *For each computable ordinal $\alpha \geq 1$, each of the following classes has a structure that is Δ_α^0 categorical but not relatively Δ_α^0 categorical.*

1. rings
2. integral domains
3. commutative semigroups
4. 2-step nilpotent groups

Proof. We start with a directed graph $\mathcal{G}^*$ that is Δ_α^0 categorical but not relatively Δ_α^0 categorical. In [12], Hirschfeldt et al showed that each of the classes of algebraic structures named in the proposition contains a structure $\mathcal{B}^*$ such that $\mathcal{G}^*$ and $\mathcal{B}^*$ satisfying their conditions—the version involving a definable quotient. Then the conditions of our Propositions 3.1 and 3.2 are satisfied. Therefore, $\mathcal{B}^*$ is Δ_α^0 categorical but not relatively Δ_α^0 categorical. $\square$

4 Fields

In [12], there is no result on fields. Just recently, Hirschfeldt, Kramer, R. Miller, and Shlapentokh showed the following [13].

Theorem 4.1 (Hirschfeldt-Kramer-Miller-Shlapentokh). *There is an algebraic field (i.e., a subfield of the algebraic closure of $\mathbb{Q}$) that is computably categorical but not relatively computably categorical.*

The proof does not involve coding a graph or a family of sets in a field in a way that can be decoded effectively. We still do not know whether this is possible in fields. By an old construction of H. Friedman and Stanley [5], we can code a graph in a field of infinite transcendence degree in a way that can be decoded in two jumps. Using this, we show that for certain limit ordinals α , there is a field that is Δ_α^0 -categorical but not relatively Δ_α^0 -categorical.

Proposition 4.2 (Fields). *For each limit ordinal $\alpha > \omega$, there is a field that is Δ_α^0 -categorical but not relatively Δ_α^0 -categorical.*

Proof. We recall some ideas from [4]. We consider families of functions $\mathcal{S} \subseteq \omega^\omega$ with enumerations G . Let $(\alpha_n)_{n \in \omega}$ be a special increasing sequence of ordinals with limit α , where $\alpha_0 > \omega$. We consider enumerations that have the form $G(i, n) = \varphi_e^{\Delta_{\alpha_n}^0}(i, n)$ where $S = \{g_i : (\forall n) g_i(n) = G(i, n)\}$. The enumerations G are “anti-Friedberg”; i.e., for each $f \in \mathcal{S}$, there are infinitely many i such that $g_i = f$. In [4], we construct a family $\mathcal{S}$ such that there is a unique anti-Friedberg enumeration of the kind we are considering, up to Δ_α^0 equivalence, and there is no Σ_α set $P \subseteq \omega^{<\omega}$ such that for each $f \in \mathcal{S}$, there exists $p \in P$ such that $p \subseteq f$ and if $f, g \in \mathcal{S}$ and $p \in P$ such that $p \subseteq f, g$, then $f = g$.

In [4], we use this family to produce a computable structure $\mathcal{A}^*$, for a finite relational language, such that $\mathcal{A}^*$ is Δ_α^0 categorical but not relatively Δ_α^0 categorical. The structure $\mathcal{A}^*$ has the following properties.

1. There is a predicate I whose elements represent indices for functions from $\mathcal{S}$. Each function is represented by infinitely many elements of I . The rest of the structure codes the values of the functions.
2. For any $\mathcal{B} \cong \mathcal{A}^*$, the family $\mathcal{S}$ has an enumeration of the form $G(i, n) = \varphi_{\alpha_n}^{\Delta_{\alpha_n}^0(\mathcal{B})}(i, n)$, where the function with index i in this enumeration is the one corresponding to the i^{th} element of $I^\mathcal{B}$ (in the ω -ordering).
3. Suppose $\mathcal{B} \cong \mathcal{A}^*$, and f is a function from $I^{\mathcal{A}^*}$ onto $I^\mathcal{B}$, such that the function corresponding to $a \in I^{\mathcal{A}^*}$ is the same as the one corresponding to $f(a) \in I^\mathcal{B}$. Then f extends to an isomorphism from $\mathcal{A}^*$ onto $\mathcal{B}$ that is Δ_α^0 relative to $\mathcal{B}$.

Using Proposition 2.3, we pass from the structure $\mathcal{A}^*$ from [4] to an undirected graph $\mathcal{G}^*$ such that $\mathcal{A}^*$ and $\mathcal{G}^*$ satisfy the conditions of our Propositions 2.1 and 2.2. There are computable Π_1 and computable Σ_1 formulas defining the universe and relations for $\mathcal{A}_{\mathcal{G}^*} \cong \mathcal{A}^*$ in $\mathcal{G}^*$.

Next, we use a transformation of H. Friedman and Stanley [5] to pass from $\mathcal{G}^*$ to a computable field $\mathcal{F}^*$ in which a copy of $\mathcal{G}^*$ is a definable quotient, given by formulas $D(x)$, $E(x, y)$ and $x \sim y$. Here is the idea of the transformation. We start with a large computable algebraically closed field $\mathcal{F}^{**}$, of characteristic 0, with a computable sequence of algebraically independent elements $(b_n)_{n \in \omega}$. Then $\mathcal{F}^*$ is the subfield of $\mathcal{F}^{**}$ generated by the following elements.

1. b_k and the elements algebraic over b_k , for $k \in \mathcal{G}^*$
2. $\sqrt{d + d'}$, where for some j, k joined by an edge in $\mathcal{G}^*$, d is inter-algebraic with b_j and d' is inter-algebraic with b_k .

The pair $\mathcal{G}^*$ and $\mathcal{F}^*$ satisfies the conditions of Proposition 3.1. Therefore, $\mathcal{F}^*$ is not relatively Δ_α^0 categorical. For Proposition 3.2, the first condition is not satisfied. The formula $D(x)$ is computable Π_2 , the formula $\sim$ is computable

Σ_1 , and the formula $E(x, y)$ giving the edge relation is finitary Σ_1 . The other important condition of Proposition 3.2 holds. There is a c.e. set of computable Σ_1 formulas defining the orbits of tuples in the field over a choice set with just one element from each $\sim$ class.

Let $\mathcal{F}$ be a computable copy of $\mathcal{F}^*$. We must show that there is a Δ_α^0 isomorphism from $\mathcal{F}^*$ onto $\mathcal{F}$. The formulas $D(x)$, $\sim$, and $E(x, y)$ give a definable quotient of $\mathcal{F}$ isomorphic to $\mathcal{G}^*$. Taking the first element of each $\sim$ -class, we get a definable copy $\mathcal{G}$ of $\mathcal{G}^*$ with universe a subset of $\mathcal{F}$. This is only Δ_3^0 —we cannot claim that it is computable. We identify $\mathcal{G}^*$ obtained by doing the same inside $\mathcal{F}^*$ —taking the first element of each $\sim$ -class. We suppose that $\mathcal{G}^*$ is computable. We identify $\mathcal{A}^*$ with the definable copy inside $\mathcal{G}^*$. This is also computable. In $\mathcal{G}$, we have the definable copy $\mathcal{A}$ of $\mathcal{A}^*$. Since $\mathcal{G}$ is Δ_3^0 , so is $\mathcal{A}$. The elements of $I^{\mathcal{A}}$ correspond to indices for the functions in $\mathcal{S}$. We have an enumeration $G(i, n)$ of $\mathcal{S}$, where $G(i, n)$ is $\Delta_{\alpha_n}^0$ relative to $\mathcal{A}$, uniformly in n . Since $\mathcal{A}$ is Δ_3^0 , we can compute a Δ_α^0 index for $\Delta_\alpha^0(\mathcal{A})$, so $G(i, n)$ has the form $\varphi_{e'}^{\Delta_{\alpha_n}^0}(i, n)$.

By the special properties of $\mathcal{S}$, we have a Δ_α^0 function f from $U^{\mathcal{A}^*}$ onto $U^{\mathcal{A}}$ such that $a \in U^{\mathcal{A}^*}$ and $f(a) \in U^{\mathcal{A}}$ represent the same function from $\mathcal{S}$. By the second property of $\mathcal{A}^*$ above, f extends to an isomorphism from $\mathcal{A}^*$ onto $\mathcal{A}$ that is Δ_α^0 relative to $\mathcal{A}$, so it is Δ_α^0 . Let g be this extension.

Next, we extend g to a Δ_α^0 isomorphism from $\mathcal{G}^*$ onto $\mathcal{G}$. We have a c.e. set of existential formulas, providing definitions of the elements of $\mathcal{G}^*$ with parameters in $\mathcal{A}^*$. Therefore, we get an extension of g to an isomorphism h from $\mathcal{G}^*$ onto $\mathcal{G}$ such that h is computable relative $\mathcal{G}$ and g . Therefore, h is Δ_α^0 .

Finally, we extend h to a Δ_α^0 isomorphism k from $\mathcal{F}^*$ onto $\mathcal{F}$. The function h maps the first element of a $\sim$ -class in $\mathcal{F}^*$ to the first element of a $\sim$ -class in $\mathcal{F}$, giving an isomorphism from $\mathcal{G}^*$ onto $\mathcal{G}$. Using the formulas that define the orbits of tuples in the field over the choice set that is the universe of the graph, we extend h to an isomorphism k from $\mathcal{F}^*$ onto $\mathcal{F}$ such that k is computable relative to $\mathcal{G}$ and h . Therefore k is Δ_α^0 . This shows that $\mathcal{F}^*$ is Δ_α^0 categorical. $\square$

5 Transferring complexity of relations

Definition 2. Let $\mathcal{A}$ be a computable structure, and let R be a relation on $\mathcal{A}$.

1. R is intrinsically Σ_α^0 on $\mathcal{A}$ if in all computable $\mathcal{B} \cong \mathcal{A}$, the image of R is Σ_α^0 .
2. R is relatively intrinsically Σ_α^0 on $\mathcal{A}$ if in all $\mathcal{B} \cong \mathcal{A}$, the image of R is Σ_α^0 relative to $\mathcal{B}$.

The notion of relatively intrinsically Σ_α^0 relation is nicer, as witnessed by the following result from [1] and [3].

Theorem 5.1 (Ash-K-Manasse-Slaman, Chisholm). *Let $\mathcal{A}$ be a computable structure and let R be a relation on $\mathcal{A}$. Then R is relatively intrinsically Σ_α^0 iff it is definable in $\mathcal{A}$ by a computable infinitary Σ_α formula, with a finite tuple of parameters.*

In [10], and [4], and [6], it is shown that for each computable ordinal α , there is a computable structure $\mathcal{A}$ with a relation R such that R is intrinsically Σ_α^0 but not relatively intrinsically Σ_α^0 . For successor ordinals, the examples involve starting with a rigid structure that is Δ_α^0 categorical but not relatively Δ_α^0 categorical. We let $\mathcal{A}$ be a structure made up of two disjoint copies of this, and we let R be the unique isomorphism between the two. For limit ordinals, the examples involve a structure built from a special family of functions, with a predicate for the set of “indices” for the functions. The relation R is a special subset of the set of indices.

Proposition 5.2. *Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there is some $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, definable in $\mathcal{B}^*$. Let $R^{\mathcal{A}^*}$ be a further relation on $\mathcal{A}^*$, invariant under automorphisms of $\mathcal{A}^*$. For simplicity, we suppose $R^{\mathcal{A}^*}$ is unary. Let $R^{\mathcal{B}^*}$ be the corresponding relation on $\mathcal{B}^*$, or on $\mathcal{A}_{\mathcal{B}^*}$.*

1. *Suppose the universe and relations of $\mathcal{A}_{\mathcal{B}^*}$ are each defined in $\mathcal{B}^*$ by a pair of formulas, one computable Σ_1 and the other computable Π_1 . If $R^{\mathcal{A}^*}$ is intrinsically Σ_α^0 on $\mathcal{A}^*$, then $R^{\mathcal{B}^*}$ is intrinsically Σ_α^0 on $\mathcal{B}^*$.*
2. *Suppose there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f is an isomorphism from $\mathcal{A}$ onto $\mathcal{A}_{\mathcal{B}}$. If $R^{\mathcal{B}^*}$ is relatively intrinsically Σ_α^0 on $\mathcal{B}^*$, then $R^{\mathcal{A}^*}$ is relatively intrinsically Σ_α^0 on $\mathcal{A}^*$.*

Proof. Proof of (1): Let g be an isomorphism from $\mathcal{A}^*$ onto $\mathcal{A}_{\mathcal{B}^*}$. Let $\mathcal{B}$ be a computable copy of $\mathcal{B}^*$, say h is an isomorphism from $\mathcal{B}^*$ onto $\mathcal{B}$. Let $R^{\mathcal{B}}$ be the h -image of $R^{\mathcal{B}^*}$. The definitions guarantee that $\mathcal{A}_{\mathcal{B}}$ is computable. Then $h \circ g$ is an isomorphism from $\mathcal{A}^*$ onto $\mathcal{A}_{\mathcal{B}}$. Then $R^{\mathcal{B}}$ is the h -image of $R^{\mathcal{B}^*}$ and the $h \circ g$ -image of $R^{\mathcal{A}^*}$, since $R^{\mathcal{A}^*}$ is invariant under isomorphisms of $\mathcal{A}^*$. Since $R^{\mathcal{A}^*}$ is intrinsically Σ_α^0 , $R^{\mathcal{B}}$ is Σ_α^0 .

Proof of (2): Let $\mathcal{A}$ be a copy of $\mathcal{A}^*$, not necessarily computable. Suppose $\Phi(\mathcal{A}) = (\mathcal{B}, f)$. Then $\mathcal{B}$ and f are computable relative to $\mathcal{A}$. Let h be an isomorphism from $\mathcal{B}^*$ onto $\mathcal{B}$. Then h maps $\mathcal{A}_{\mathcal{B}^*}$ isomorphically onto $\mathcal{A}_{\mathcal{B}}$. The h -image of $R^{\mathcal{B}^*}$ is $R^{\mathcal{B}}$. Therefore, $R^{\mathcal{B}}$ is Σ_α^0 relative to $\mathcal{A}$. Since $R^{\mathcal{A}^*}$ is invariant under automorphisms of $\mathcal{A}^*$, we have $x \in R^{\mathcal{A}^*}$ iff $(\exists y) (f(x) = y \ \& \ y \in R^{\mathcal{B}})$. The right-hand side is Σ_α^0 relative to $\mathcal{A}$. □

As with the categoricity results, a similar result as the one in the proposition above can be obtained in the case that a copy of the structure $\mathcal{A}$ can be found as a quotient in another structure $\mathcal{B}$.

Proposition 5.3. *Let $\mathcal{A}^*, \mathcal{B}^*$ be computable structures for finite relational languages. Suppose there exists $\mathcal{A}_{\mathcal{B}^*} \cong \mathcal{A}^*$, a definable quotient in $\mathcal{B}^*$, using formulas invariant under automorphisms of $\mathcal{B}^*$. If $\mathcal{B} \cong \mathcal{B}^*$, we write $\mathcal{A}_{\mathcal{B}}$ for the corresponding definable quotient of $\mathcal{B}$. Let $R^{\mathcal{A}^*}$ be a further relation on $\mathcal{A}^*$, invariant under automorphisms of $\mathcal{A}^*$. For simplicity, we suppose $R^{\mathcal{A}^*}$ is unary. Let $R^{\mathcal{B}^*}$ consist of the elements of $\mathcal{B}^*$ belonging to the equivalence classes that correspond to elements of $R^{\mathcal{A}^*}$.*

1. *Suppose the set, equivalence relation, denoted by $\sim$, and basic relations that give $\mathcal{A}^{\mathcal{B}^*}$ are definable by both computable Σ_1 and computable Π_1 formulas. If $R^{\mathcal{A}^*}$ is intrinsically Σ_α^0 on $\mathcal{A}^*$, then $R^{\mathcal{B}^*}$ is intrinsically Σ_α^0 on $\mathcal{B}^*$.*
2. *Suppose there is a Turing operator Φ , defined on all $\mathcal{A} \cong \mathcal{A}^*$, such that $\Phi(\mathcal{A}) = (\mathcal{B}, f)$, where $\mathcal{B} \cong \mathcal{B}^*$ and f maps $\mathcal{A}$, 1 – 1 into $\mathcal{B}$, so that $\text{ran}(f)$ has just one element of each equivalence class, and f induces an isomorphism from $\mathcal{A}$ onto the quotient structure $\mathcal{B}_{\mathcal{A}}$. If $R^{\mathcal{B}^*}$ is relatively intrinsically Σ_α^0 on $\mathcal{B}^*$, then $R^{\mathcal{A}^*}$ is relatively intrinsically Σ_α^0 on $\mathcal{A}^*$.*

Proof. Proof of (1): Let $\mathcal{B}$ be a computable copy of $\mathcal{B}^*$. Taking the first element of each $\sim$ -class, we get a computable copy $\mathcal{A}'$ of $\mathcal{A}^*$ with universe a subset of $\mathcal{B}$. The relation $R^{\mathcal{A}'}$ corresponding to $R^{\mathcal{A}^*}$ is Σ_α^0 . We have $x \in R^{\mathcal{B}}$ iff $(\exists y)(y \sim x \text{ \& } y \in R^{\mathcal{A}'})$. Therefore, $R^{\mathcal{B}}$ is Σ_α^0 .

Proof of (2): Let $\mathcal{A}$ be a copy of $\mathcal{A}^*$, not necessarily computable. Let $\Phi(\mathcal{A}) = (\mathcal{B}, f)$. Then $\mathcal{B}$ and f are computable relative to $\mathcal{A}$. Since $R^{\mathcal{B}^*}$ is relatively intrinsically Σ_α^0 , $R^{\mathcal{B}}$ is Σ_α^0 relative to $\mathcal{A}$. We have $x \in R^{\mathcal{A}}$ iff $(\exists y)[f(x) = y \text{ \& } y \in R^{\mathcal{B}}]$. Therefore, $R^{\mathcal{A}}$ is Σ_α^0 relative to $\mathcal{A}$. □

6 Questions

Definition 3. *A computable structure $\mathcal{A}$ is hyperarithmetically categorical if for all computable copies, there is a hyperarithmetical isomorphism, and it relatively hyperarithmetically categorical if for all copies $\mathcal{B}$, there is an isomorphism that is hyperarithmetical relative to $\mathcal{B}$.*

Problem 1. *If $\mathcal{A}$ is hyperarithmetically categorical, must it be relatively hyperarithmetically categorical?*

If $\mathcal{A}$ is rigid, then this question has a positive answer. Consider the structure $\mathcal{A}^*$ that is the cardinal sum of two copies of $\mathcal{A}$. Let R be the relation on $\mathcal{A}^*$ that is the isomorphism between the two copies of $\mathcal{A}$. If $\mathcal{A}$ is relatively hyperarithmetically categorical, then R is relatively intrinsically hyperarithmetical on $\mathcal{A}^*$. By results of I. Soskov [18, 19], reworked in [11], R is intrinsically hyperarithmetical on $\mathcal{A}$ then it is relatively intrinsically hyperarithmetical on $\mathcal{A}$.

Recently, Kach et al have shown that for each computable ordinal α , there is a computably categorical structure $\mathcal{A}$ that is not relatively Δ_α^0 categorical. They showed, further, that the set of computable indices for computably categorical structures is m -complete Π_1^1 . This suggests the following question.

Problem 2. *Is there a computably categorical structure $\mathcal{A}$ that is not relatively hyperarithmetically categorical?*

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